

Computer Vision

Major Stages of Visual Data Handling Pipeline

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Overview

- Computer Vision enables computers to interpret images.
- Pipeline: Acquisition → Preprocessing → Feature Extraction → Recognition → Output.
- Used in healthcare, robotics, surveillance, and autonomous driving.
- The Computer Vision pipeline transforms a real-world scene into meaningful digital information through a series of processing stages.

Image Acquisition

- Cameras or sensors capture real-world scenes.
- Analog signals are converted into digital images.
- Lighting, resolution, focus, and exposure affect image quality.
- Better acquisition improves system performance.

Preprocessing & Feature Extraction

- Removes noise and enhances image quality.
- Includes filtering, resizing, normalization, and contrast enhancement.
- Extracts edges, corners, textures, and shapes.
- Features are inputs for recognition algorithms.

Analysis & Recognition

- Machine Learning and Deep Learning analyze extracted features.
- Performs object detection, classification, segmentation, and tracking.
- Produces meaningful outputs for decision making.

Applications

- Face recognition
- Medical image analysis
- Autonomous vehicles
- Industrial inspection
- Smart surveillance

Conclusion

- Every stage is essential for accurate computer vision.
- High-quality input leads to better predictions.
- Computer Vision converts real-world images into meaningful digital information through a structured processing pipeline.
- High-quality image acquisition and effective preprocessing significantly enhance system performance